NETWORKING

Efficient Scale-up Enabled by Active Copper Cables

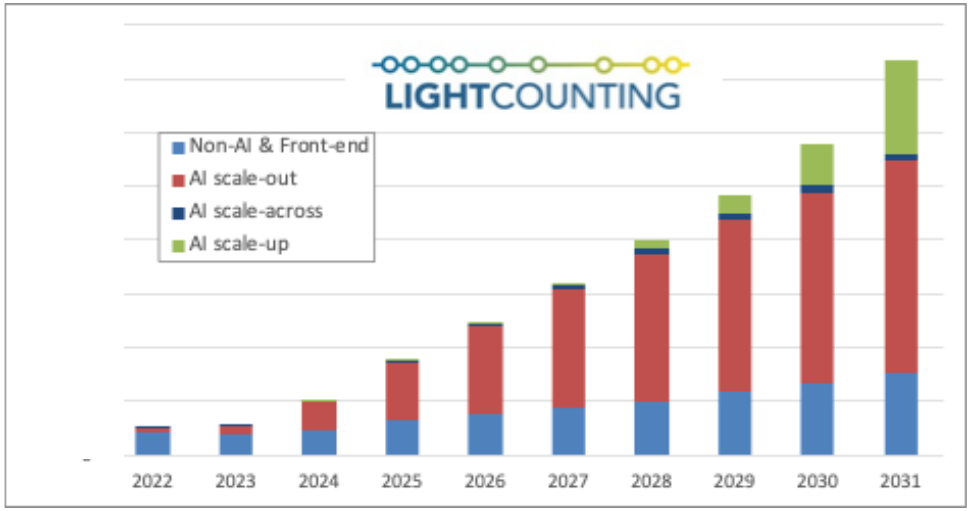
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Product and Technology Marketing, Ciena

OCP BRONZE™ Member

AI driving networking innovation

- Optical Ethernet ports to eclipse 350M ports in 5 years
- Data rates & scale are driving network technology inflection points as people seek efficient solutions at massive scale
  - Front panel pluggables → CPO/NPO
  - Coherent lite optics → 10km + hi loss intra-DC
- Scale-up interconnect
  - Multi-rack scale up requires optics
  - At 200Gbps copper is still a good choice for intra-rack and rack-to-rack applications
- Passive copper value proposition
  - Readily available
  - Highly reliable
  - Easy to use – plug & play
  - Low cost
  - But….



Source: LightCounting "Cloud Data Center Optics", July 2026

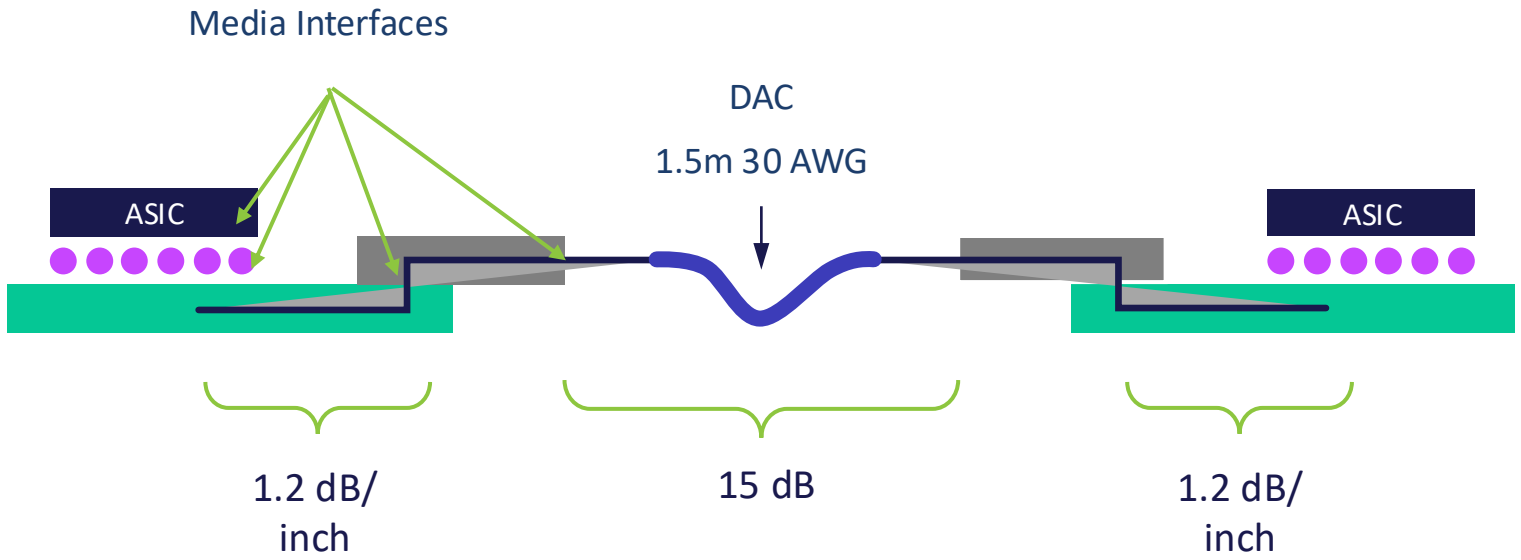


Limitations of DAC

IEEE PMD	IEEE Standard	Signaling per Lane	Target Reach
50GBASE-CR	IEEE 802.3cd	1 x 50G PAM4	3 meters
100GBASE-CR1	IEEE 802.3ck	1 x 100G PAM4	2 meters – 33% decrease
200GBASE-CR1	IEEE 802.3dj	1 x 200G PAM4	1 meter – 50% decrease

Example 40dB channel at 200Gbps

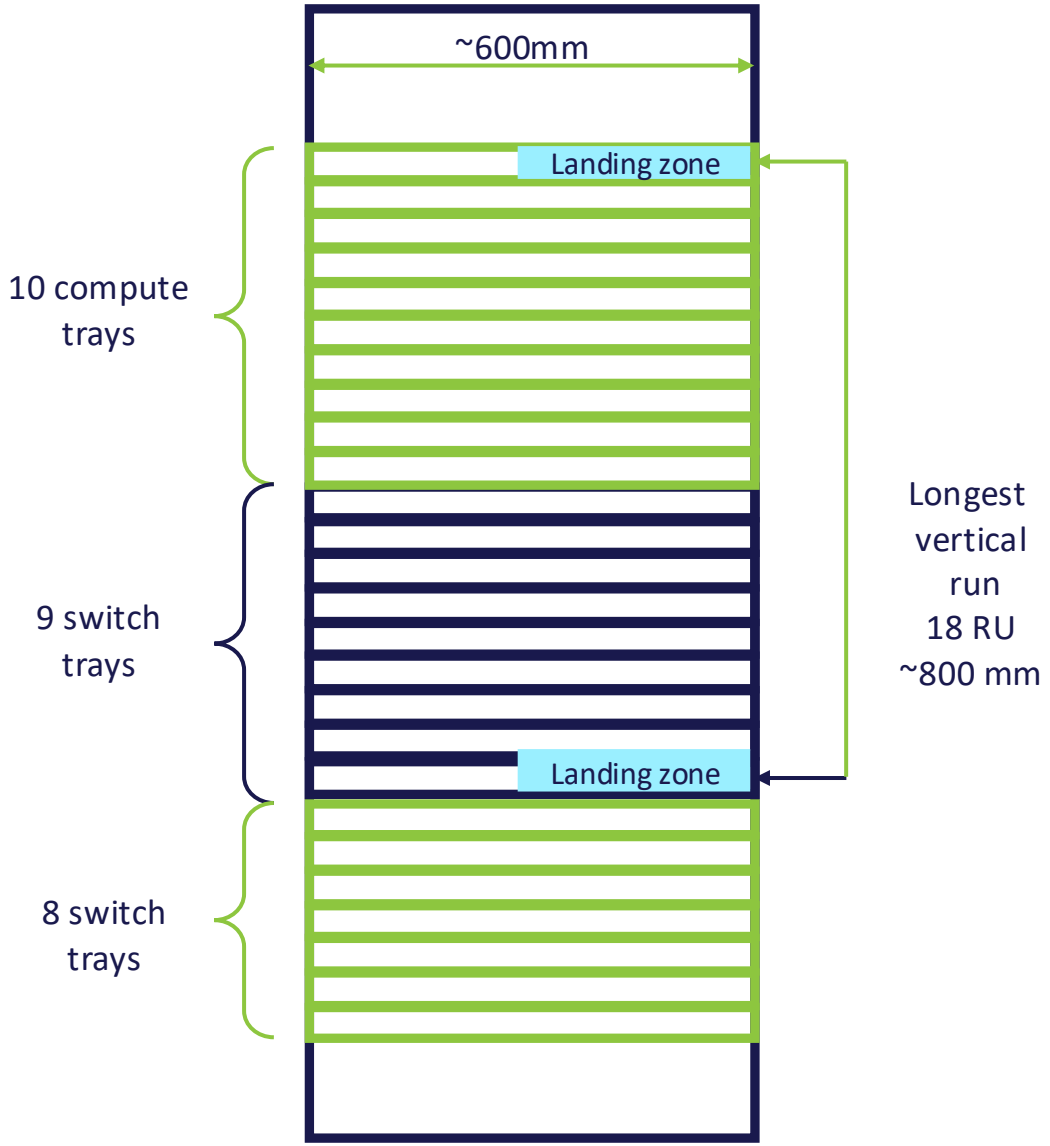
Media	Loss
1.5m Cable	16dB
Media Interfaces	2 x 5 = 10dB
Remaining for PCB	14 / 1.2 = ~12 in ***6" per side***



1.5 meter cable loss budget leaves ~6" of low-loss PCB trace

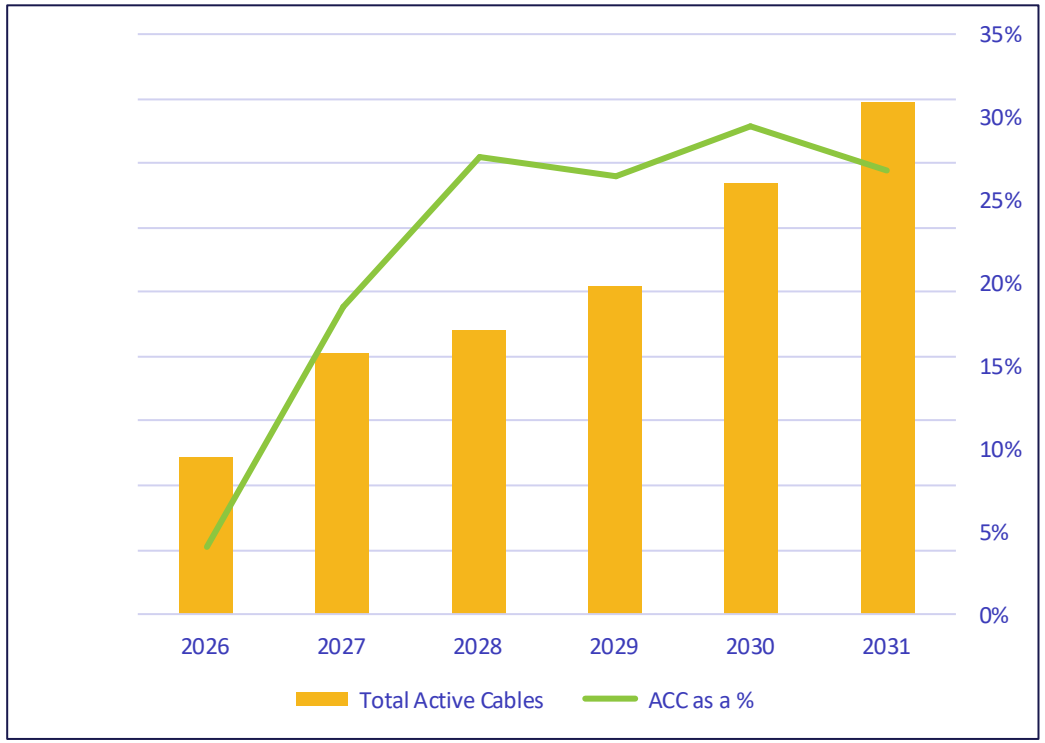
DAC hitting limits

- A fully engineered AI rack with an advanced cabled backplane pushes DAC technology its limits
  - Note only ~350mm available for horizontal interconnect
  - System takes 40dB EQ to its limit
- Interconnect becomes more complex with scale-up networks that are both intra and inter rack
  - Increased number of cables, types and routes
  - Optical rack to rack
  - Copper intra-rack
- Options when DAC runs out of steam
  - Optical
  - Active copper



Rise of active copper

- As the reach of passive cables declines, active cables become more prevalent
- Active cable extend the reach of cables using boost elements
- Active copper retains much of DAC value prop
  - Reliable
  - Available
  - Lower cost
- Two primary types
  - Active electrical cables – AECs
  - Active copper cables – ACCs
- Active cable forecast to 10’s of millions in 5 years
  - ACC ~30% of active market



Source: LightCounting “Cloud Data Center Optics”, July 2026

Moving from passive to active copper

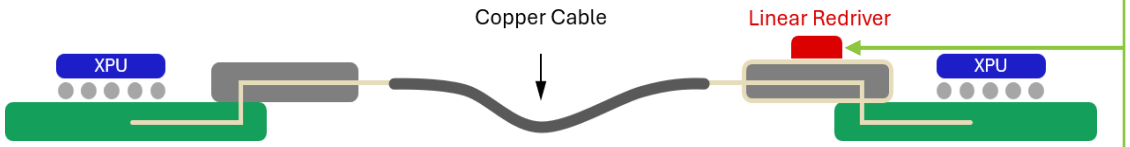
How to make active copper retain look and feel of DAC

Minimize Power

Avoid Complexity:
- Configuration & software
- # of power supplies
- MSAP PCB for KGD

1.6T Active Copper Cable (ACC)

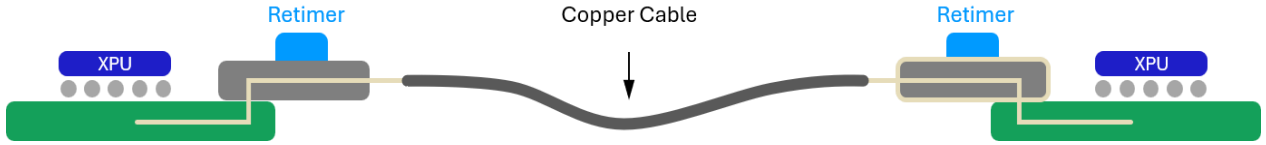
ACC-MSA





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Nitro 2004
Linear Redriver

1.6T Active Electrical Cable (AEC)



8 x 200G
Active Cable Comparison

	Length	Power per end	Added Latency	Power Supplies	Assembly	Cost	Control plane
ACC	3.5 m	1.5pJ /bit	Zero	3.3V No DC2DC	BGA	$	Plug and play
AEC	4.5 m	10pJ/ bit	~2 x 35ns	4+ rails	BGA / KGD	$$$$	Full API library

ACC where you can. AEC where you must.

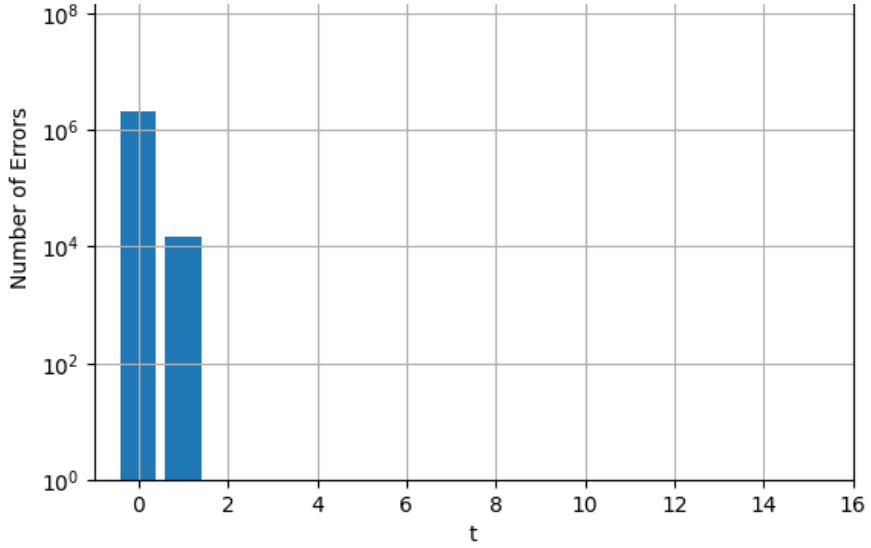
Active Copper Cable MSA

- MSA announced Feb 23rd 2026
- Goal – accelerate ACC adoption through the creation of competitive, standards-based supply chain
  - Make ACCs standard products that can be multi-sourced
- 16 member companies – more applications in process
- First plugfest held Aug 31st to Sep 4th 2026
  - Characterize cables to aid in specification development
  - Measure BER performance of cables
  - 8 participating companies
  - 32 cables tested
  - Data currently being anonymized

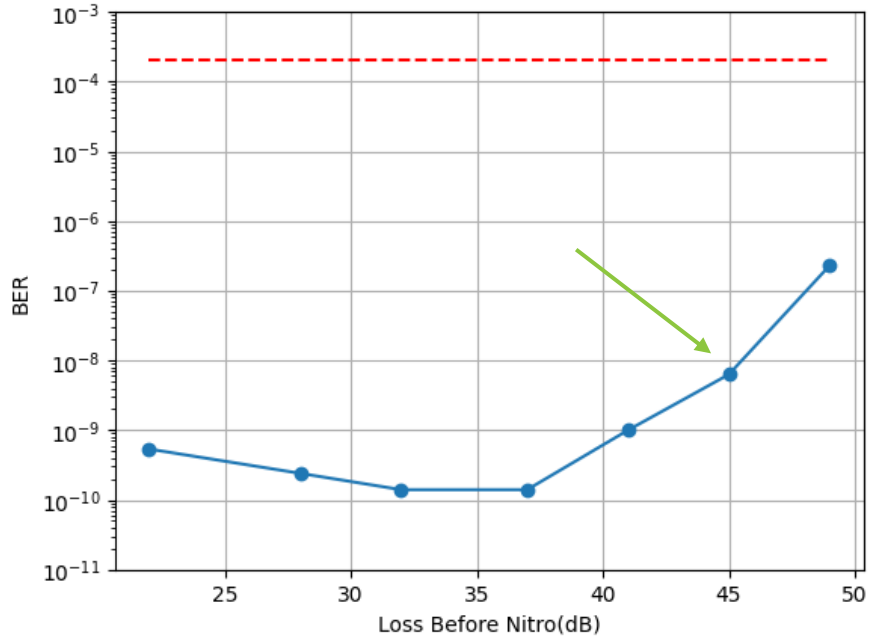
ACC MSA Members

Example of ACC performance

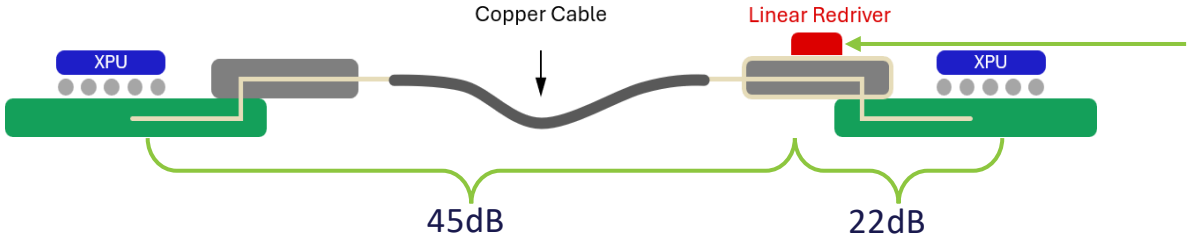
45 dB loss before, 22 dB Loss after Nitro
67 dB Total Loss Channel

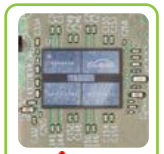


22 dB Loss After Nitro



1.6T Active Copper Cable (ACC)





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Nitro 2004 Linear Redriver

Conclusion

- DACs may be length challenged for some scale-up network implementations, even intra-rack
- Active cables offer extended reach
  - AECs maximize copper reach with some expense
  - ACC's are cost effective, plug and play alternative that retains many of the value propositions of DAC, while extending 200Gbps cables by over 1m
- The ACC MSA is working to create a standard with that will enable an ACC supply chain
  - Resilient sourcing from world's largest cable vendors
  - Ensure that ACCs behave as standard products that can be multi-sourced






Thank
You!
OCP
GLOBAL
SUMMIT
October 12-15, 2026
San Jose, California